

AWS Client VPN ile Güvenli Altyapı Erişimi

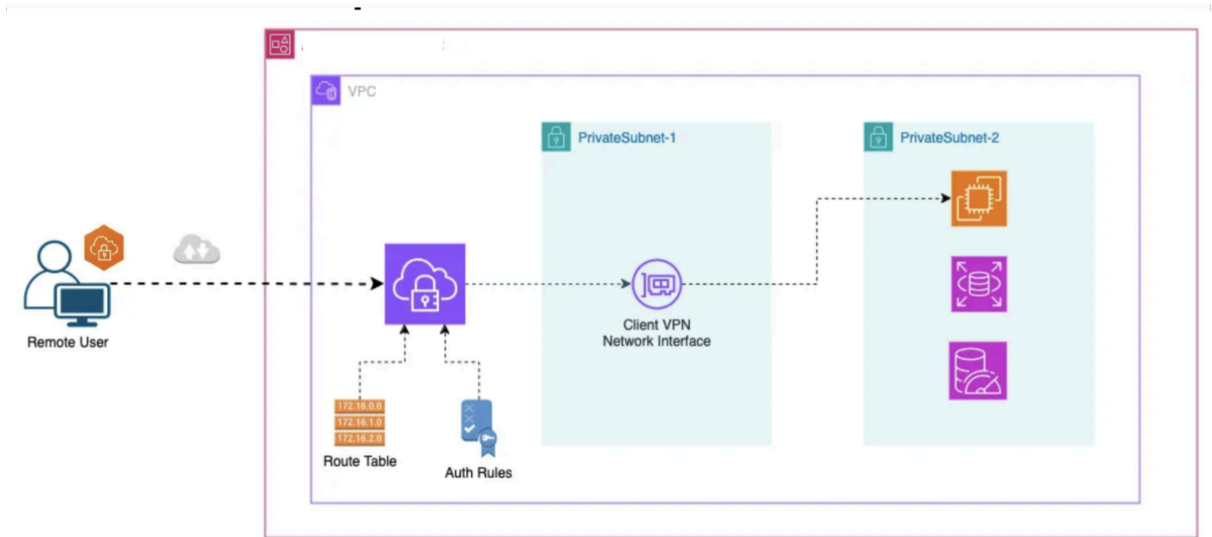
Giriş: AWS Client VPN Nedir?

Kurumsal ortamlarda geliştiricilerin, DevOps mühendislerinin veya yöneticilerin AWS üzerindeki **özel (private) kaynaklara** güvenli şekilde erişmesi gerekir. Örneğin; **private subnet'te çalışan bir EC2, veritabanı**, ya da sadece içeriden erişilebilen servisler.

Bunun için AWS, **Client VPN** adında tamamen yönetilen bir VPN çözümü sunar. AWS Client VPN şunları sağlar:

-  **Güvenli uzaktan erişim** → Private subnet içindeki kaynaklara dışarıdan bağlanabilirsin.
-  **Mutual TLS (sertifika tabanlı kimlik doğrulama)** → Hem sunucu hem de istemci tarafında sertifikalar kullanılır.
-  **Tamamen AWS tarafından yönetilen** → EC2'de OpenVPN sunucusu kurmanıza gerek yok.
-  **Esnek entegrasyon** → IAM, Active Directory ya da mutual TLS ile kimlik doğrulama yapılabilir.
-  **Yüksek erişilebilirlik** → Birden fazla AZ'de subnet association ile HA sağlanabilir.

Yani AWS Client VPN sayesinde, ek bir OpenVPN sunucusu işletmeden, doğrudan AWS tarafından yönetilen bir VPN altyapısına sahip oluyorsun.



Lab Senaryosu

1. Sertifika otoritesi (CA), server ve client sertifikaları oluşturacağız.
2. Sertifikaları AWS Certificate Manager (ACM)'e import edeceğiz.
3. AWS Client VPN Endpoint kuracağız.
4. Target subnet association ve authorization rule ekleyeceğiz.
5. Client config dosyası (.ovpn) ile bağlantı test edeceğiz.
6. VPN üzerinden private EC2'ye erişim sağlayacağız.

1 Ortam Hazırlığı

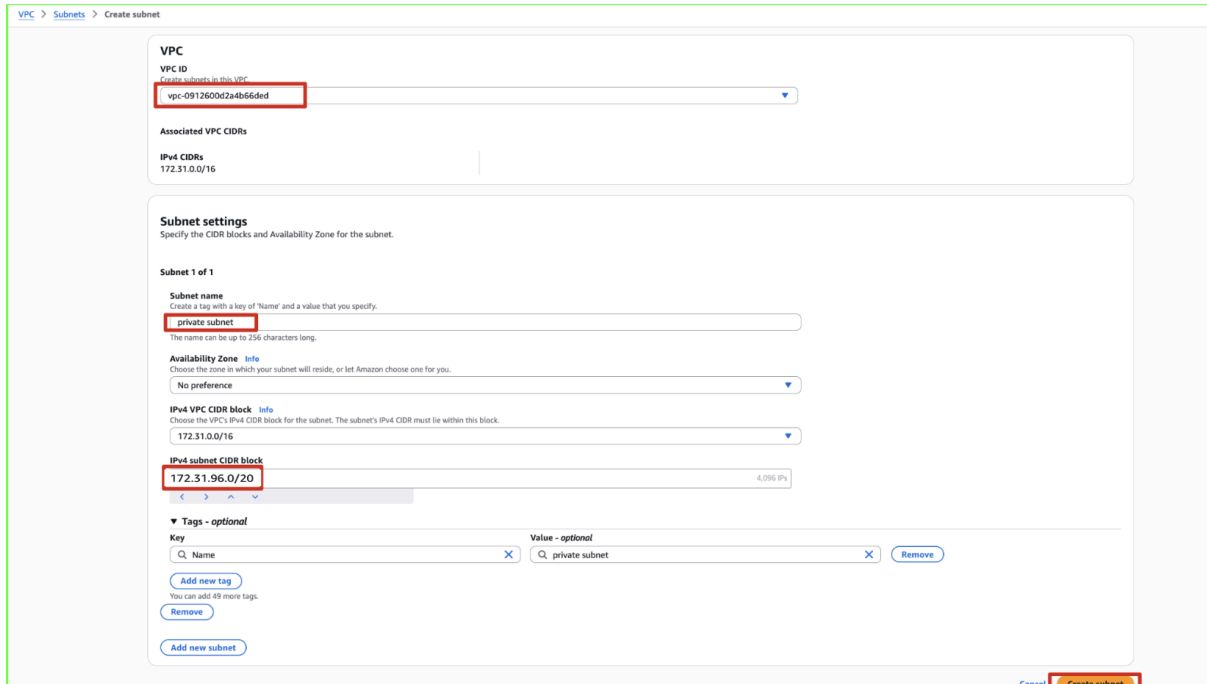
- AWS CLI kurulu ve configure edilmiş olmalı (**aws configure**)
- OpenSSL veya Easy-RSA kurulu olmalı (sertifika üretmek için)
- ACM, VPC, Subnet, Security Group yönetim yetkilerin olmalı
- Kurulum yapılacak **region** baştan seçilmeli (ör. **us-east-1** veya **eu-central-1**)

1.1 IP PLANI

- **VPC (Default VPC):** **172.31.0.0/16** (AWS'in varsayılanı)
- **Private Subnet:** **172.31.96.0/20** (yeni oluşturacağız)
- **Client VPN "Client CIDR":** **172.16.0.0/20**

1.2 IP Private Subnet ve EC2-test sunucu kurulumu

Default VPC içinde yeni bir **private subnet** aç: **172.31.96.0/20**



VPC > Subnets > Create subnet

VPC

VPC ID
Create subnets in this VPC
vpc-0912600d2a4b66ded

Associated VPC CIDRs

IPv4 CIDRs
172.31.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private subnet
The name can be up to 256 characters long.

Availability Zone
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.31.0.0/16

IPv4 subnet CIDR block
172.31.96.0/20
4,096 IPs

Tags - optional

Key
Name

Value - optional
private subnet

Add new tag
You can add 49 more tags.

Remove

Add new subnet

Cancel Create subnet

EC2-Test security group oluřturalım

EC2→Security Groups→Create security group

Security group name:ec2-test

Description: ec2-test

VPC :Default

Inbound rules:

Rules: SSH → Source: Anywhere (0.0.0.0/0)

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details [Info](#)

Security group name [Info](#)

ec2-test

Name cannot be edited after creation.

Description [Info](#)

ec2-test

VPC [Info](#)

vpc-09126002a4b66ded

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional
SSH	TCP	22	Custom	172.16.0.0/22

[Add rule](#)

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Create security group](#)

EC2-Test sunucusu oluřturalım

- Bölge: **us-east-1** (N. Virginia).
- **Services** → **EC2** → **Instances** → **Launch instances**.
- **Name and tags:**
 - **Name:** EC2-test
- **Application and OS Images (AMI):**
 - **Quick Start** → **Amazon Linux 2023 (kernel-6.1)**

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

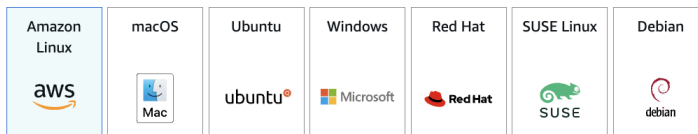
[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs

[Quick Start](#)



[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-08982f1c5bf93d976 (64-bit (x86), uefi-preferred) / ami-039f81f5ce6752b10 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250915.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username ⓘ
64-bit (x86) ▼	uefi-preferred	ami-08982f1c5bf93d976	2025-09-10	ec2-user

Verified provider

● Instance type: **t2.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

● Key Pair (login):

- **Create new key pair** → Key pair name: **ec2-ssh-key**
- **Type:** RSA, **File format:** **.pem** → Create ve seç.

Create key pair



Key pair name

Key pairs allow you to connect to your instance securely.

ec2-ssh-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type



RSA

RSA encrypted private and public key pair



ED25519

ED25519 encrypted private and public key pair

Private key file format



.pem

For use with OpenSSH



.ppk

For use with PuTTY



When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

Key dosyasını bilgisayarına kaydet

● Network settings → Edit:

- VPC: default
- Subnet: varsayılan (Private subnet)
- Auto-assign public IP: disable
- Firewall (SG): Create a new security group

■ Security group name: ec2-test

■ Description: ec2-test

■ Rules:

1. SSH → Source: Anywhere (0.0.0.0/0)

▼ Network settings Info

VPC - required Info

vpc-0912600d2a4b66ded
172.31.0.0/16

(default)



Subnet Info

subnet-0d9b5f5b35d6b5d02

private subnet

VPC: vpc-0912600d2a4b66ded Owner: 922897610805 Availability Zone: us-east-1a (use1-az4)
Zone type: Availability Zone IP addresses available: 4089 CIDR: 172.31.96.0/20



Create new subnet

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups Info

Select security groups

ec2-test sg-016ca04f3ef1c4ba0
VPC: vpc-0912600d2a4b66ded



Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Launch instance butonuna tıklayarak sunucuyu oluşturun.

The screenshot shows the AWS Management Console 'Instances' page. At the top, there's a search bar and a table of instances. The first instance, 'ec2-test', is highlighted with a red box. Below the table, the 'Details' tab is selected for instance 'i-Oc8839743db25af7e'. In the 'Instance summary' section, the 'Private IPv4 addresses' field is highlighted with a red box, showing the address '172.31.98.215'.

- EC2'nun IPv4 Private IP adresini not et.

2) Sertifika Altyapısı (CA, Server, Client)

AWS Client VPN, **Mutual TLS Authentication** kullanır:

- Bir **Server certificate** (endpoint kimliği için)
- Bir **Client certificate** (kullanıcı kimliği için)
- İkisini de imzalayan bir **CA (Root CA)**

Sertifikaları easy-rsa ile oluşturabiliriz:

2.1 Easy-RSA indir & başlat

```
git clone https://github.com/OpenVPN/easy-rsa.git
cd easy-rsa/easyrsa3
# PKI başlat
./easyrsa init-pki
```

```
(base) aws-client-vpn ➡ git clone https://github.com/OpenVPN/easy-rsa.git
Cloning into 'easy-rsa'...
remote: Enumerating objects: 8674, done.
remote: Counting objects: 100% (188/188), done.
remote: Compressing objects: 100% (69/69), done.
remote: Total 8674 (delta 120), reused 121 (delta 117), pack-reused 8486 (from 3)
Receiving objects: 100% (8674/8674), 62.29 MiB | 4.50 MiB/s, done.
Resolving deltas: 100% (4126/4126), done.
(base) aws-client-vpn ➡ cd easy-rsa/easyrsa3
(base) easyrsa3 ➡ ./easyrsa init-pki

Notice
-----
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki

Using Easy-RSA configuration:
* undefined
```

2.2 Root CA oluşturun

```
# Root CA oluşturun
./easysrsa build-ca nopass
```

```
(base) easysrsa3 ➡ ./easysrsa build-ca nopass
.....+++++
.....+++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]: devopsatolyesi.com

Notice
-----
CA creation complete. Your new CA certificate is at:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/ca.crt

Build-ca completed successfully.
```

2.3 Server sertifikası oluşturun

```
# Server sertifikası
./easysrsa build-server-full devopsatolyesi.com nopass
```

```
(base) easysrsa3 ➡ ./easysrsa build-server-full devopsatolyesi.com nopass
Generating a RSA private key
.....+++++
.....+++++
writing new private key to '/Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/e536692c/temp.03'
-----
Notice
-----
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/reqs/devopsatolyesi.com.req
* key: /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/private/devopsatolyesi.com.key

You are about to sign the following certificate:

  Requested CN:      'devopsatolyesi.com'
  Requested type:    'server'
  Valid for:         '825' days

subject=
  commonName          = devopsatolyesi.com

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/a482cf7d/temp.02
```

```
Using configuration from /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/8e4bc52f/temp.02
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName      :ASN.1 12:'server'
Certificate is to be certified until Dec 31 11:24:52 2027 GMT (825 days)

Write out database with 1 new entries
Data Base Updated

WARNING
-----
INCOMPLETE Inline file created:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/inline/private/server.in
line

Notice
-----
Certificate created at:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easysrsa3/pki/issued/server.crt
```

2.4 Client sertifikası oluştur

```
# Client sertifikası (ör: client1)
./easyrsa build-client-full client1.devopsatolyesi.com nopass
```

```
(base) easyrsa3
(base) easyrsa3 ./easyrsa build-client-full client1.devopsatolyesi.com nopass
Generating a RSA private key
.....+++++
.....+++++
writing new private key to '/Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/ea7794f8/temp.03'
-----

Notice
-----
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/reqs/client1.devopsatolyesi.com.req
* key: /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/private/client1.devopsatolyesi.com.key

You are about to sign the following certificate:

Requested CN:      'client1.devopsatolyesi.com'
Requested type:    'client'
Valid for:         '825' days

subject=
  commonName              = client1.devopsatolyesi.com

Type the word 'yes' to continue, or any other input to abort.
Confirm requested details: yes

Using configuration from /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/f
```

```
04ff1e5/temp.02
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName      :ASN.1 12:'client1.devopsatolyesi.com'
Certificate is to be certified until Dec 31 11:44:57 2027 GMT (825 days)

Write out database with 1 new entries
Data Base Updated

WARNING
=====
INCOMPLETE Inline file created:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/inline/private/client1.devopsatolyesi.com.inline

Notice
-----
Certificate created at:
* /Users/hakan/sistem-network-atolyesi/7-ekim/aws-client-vpn/easy-rsa/easyrsa3/pki/issued/client1.devopsatolyesi.com.crt
```

Çıktı dosyaları:

- **pki/ca.crt** → Root CA
- **pki/private/ca.key**
- **pki/issued/devopsatolyesi.com.crt** → Server sertifika
- **pki/private/devopsatolyesi.com.key** → Server private key
- **pki/issued/client1.devopsatolyesi.com.crt** → Client sertifika
- **pki/private/client1.devopsatolyesi.com.key** → Client private key

👉 Bu aşamada 3 tip dosya seti hazır:

- Root CA: **ca.crt**
- Server: **devopsatolyesi.com.crt + devopsatolyesi.com.key**
- Client: **client1.devopsatolyesi.com.crt + client1.devopsatolyesi.com.key**

3 Sertifikaları AWS ACM'e Import Et

AWS Client VPN endpoint yalnızca ACM içindeki sertifikaları kullanabilir. Bu yüzden hem server hem client sertifikalarını yükleyeceğiz.

3.1 Server certificate import

```
aws acm import-certificate \  
  --certificate fileb://pki/issued/devopsatolyesi.com.crt \  
  --private-key fileb://pki/private/devopsatolyesi.com.key \  
  --certificate-chain fileb://pki/ca.crt
```

```
(base) easyrsa3 ➦ aws acm import-certificate --certificate fileb://pki/issued/devopsatolyesi.com.crt --private-key fileb://pki/private/devopsatolyesi.com.key --certificate-chain fileb://pki/ca.crt  
{  
  "CertificateArn": "arn:aws:acm:us-east-1:922897610805:certificate/96e607c9-0c23-4b22-87c9-9c701d7c2c4f"  
}  
(base) easyrsa3 ➦
```

3.2 Client certificate import

```
aws acm import-certificate \  
  --certificate fileb://pki/issued/client1.devopsatolyesi.com.crt \  
  --private-key fileb://pki/private/client1.devopsatolyesi.com.key \  
  --certificate-chain fileb://pki/ca.crt
```

```
(base) easyrsa3 ➦ aws acm import-certificate \  
  --certificate fileb://pki/issued/client1.devopsatolyesi.com.crt \  
  --private-key fileb://pki/private/client1.devopsatolyesi.com.key \  
  --certificate-chain fileb://pki/ca.crt  
{  
  "CertificateArn": "arn:aws:acm:us-east-1:922897610805:certificate/594f23d8-be51-401a-b499-cda317c435bd"  
}  
(base) easyrsa3 ➦
```

💡 Bu ARN'leri kaydet → birazdan Client VPN Endpoint oluştururken kullanacağız.

Certificates (2)				
🔄 More actions ▼ Manage expiry events Import Request				
< 1 > ⚙️				
☐	Certificate ID	Domain name ▼	Type ▼	Status ▼
☐	594f23d8-be51-401a-b499-cda317c435bd	client1.devopsatolyesi.com	Imported	🟢 Issued
☐	96e607c9-0c23-4b22-87c9-9c701d7c2c4f	devopsatolyesi.com	Imported	🟢 Issued

3 Client VPN Endpoint Oluřturma

AWS Console hesabınız le ile giriř yaptıktan sonra

VPC → Virtual private network (VPN) → Client VPN Endpoint → Create client VPN endpoint

- **Name tag:** client-vpn
- **Description:** client-vpn
- **Endpoint IP address type:** IPv4
- **Traffic IP address type:** IPv4
- **Client IPv4 CIDR** → 172.16.0.0/20 (VPN kullanıcılarına atanacak IP aralığı)

☰ VPC > Client VPN endpoints > Create client VPN endpoint

Create client VPN endpoint [Info](#)

Create a client VPN endpoint to enable access to networks over a TLS VPN session.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

client-vpn

Name must be 255 characters or less in length.

Description - optional
A brief description of the client VPN endpoint.

client-vpn

Endpoint IP address type
The type of connection between the VPN client and server endpoint.

☒ IPv4

☐ IPv6

☐ Dual stack

Traffic IP address type
The type of traffic passing through the VPN tunnel.

☒ IPv4

☐ IPv6

☐ Dual stack

Client IPv4 CIDR [Info](#)
The IP address range, in CIDR notation, from which client IP addresses are allocated when the traffic type is IPv4.

172.16.0.0/20

CIDR block cannot be larger than /12 or smaller than /22.

- **Server certificate ARN** → ACM'deki server cert

Authentication information [Info](#)

Server certificate ARN
The server certificate must be provisioned with or imported into AWS Certificate Manager (ACM).

Select certificate

Q

arn:aws:acm:us-east-1:922897610805:certificate/96e607c9-0c23-4b22-87c9-9c701d7c2c4f
devopsatolyesi.com

arn:aws:acm:us-east-1:922897610805:certificate/594f23d8-be51-401a-b499-cda317c435bd
client1.devopsatolyesi.com

- **Authentication** → *Mutual authentication* + Client Root CA ARN

Authentication information [Info](#)

Server certificate ARN

The server certificate must be provisioned with or imported into AWS Certificate Manager (ACM).

arn:aws:acm:us-east-1:922897610805:certificate/96e607c9-0c23-4b22-87c9-9c701d7c2c4f

Q

arn:aws:acm:us-east-1:922897610805:certificate/96e607c9-0c23-4b22-87c9-9c701d7c2c4f
devopsatolyesi.com

arn:aws:acm:us-east-1:922897610805:certificate/594f23d8-be51-401a-b499-cda317c435bd
client1.devopsatolyesi.com

Select certificate

Authentication information [Info](#)

Server certificate ARN

The server certificate must be provisioned with or imported into AWS Certificate Manager (ACM).

arn:aws:acm:us-east-1:922897610805:certificate/96e607c9-0c23-4b22-87c9-9c701d7c2c4f

Authentication options

Choose one or a combination of authentication methods to use.

- ☒ Use mutual authentication
☐ Use user-based authentication

Client certificate ARN [Info](#)

arn:aws:acm:us-east-1:922897610805:certificate/594f23d8-be51-401a-b499-cda317c435bd

Connection logging [Info](#)

- ☐ Enable log details on client connections
Use connection logs to have forensics on when clients attempted to connect.

Client connect handler [Info](#)

- ☐ Enable client connect handler

- Transport protocol → UDP (daha hızlı)

DNS configuration - optional

There is no default DNS server. The DNS server IP must match the configured traffic type. When configured for dual-stack traffic, the system will choose servers with IP address types that match th

DNS server 1 IP address

192.168.0.2

DNS server 2 IP address

169.254.169.253

DNS server 3 IP address

2a05:d018:fff4:7100::2

DNS server 4 IP address

fd00:ec2::253

Other parameters - optional

Transport protocol [Info](#)

Transport protocol used by the TLS sessions.

- ☒ UDP
☐ TCP
☐ Enable split-tunnel [Info](#)

- **Split-tunnel** → Aktif (sadece VPC trafiği VPN'den geçer)

☒ Enable split-tunnel [info](#)

VPC ID

vpc-0912600d2a4b66ded

Security group IDs

Security groups to be applied to the endpoint.

Select security groups

VPN port

AWS client VPN supports ports 443 and 1194 for both TCP and UDP.

443

☒ Enable self-service portal [info](#)

Session timeout hours [info](#)

24

☒ Disconnect on session timeout [info](#)

☒ Enable client login banner [info](#)

☒ Client route enforcement [info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Add new tag

You can add 49 more tags.

Value - optional

Q client-vpn

Remove

Cancel

Create client VPN endpoint

Client VPN endpoints (1/1) [info](#)

Find client VPN by attribute or tag

Name	Client VPN endpoint ID	State	Client CIDR
☒ client-vpn	cvpn-endpoint-0a06f8f0926c4e211	☒ Available	172.16.0.0/20

4 Target Network Association

- Endpoint oluşturulduktan sonra “Associate Target Network” seçeneğine gir.

Client VPN endpoints (1/1) [info](#)

Name ↗

Client VPN endpoint ID

State

Client CIDR

client-vpn

cvpn-endpoint-07897e208651f5583

Pending-associate

172.16.0.0/19

cvpn-endpoint-07897e208651f5583 / client-vpn

Details

Target network associations

Security groups

Authorization rules

Route table

Connections

Tags

Target network associations [info](#)

Association ID

State

Network ID

Security groups

Endpoint ID

No target networks

Disassociate

Associate target network

- Private subnet seç.

VPC > Client VPN endpoints > cvpn-endpoint-0a06f8f0926c4e211 > Associate target network

Associate target network Info

A target network is a subnet in a VPC. You associate a subnet in an Availability Zone to the client VPN endpoint. You can associate one subnet per Availability Zone. You can associate subnets in one VPC to a client VPN endpoint.

Details

Client VPN endpoint ID
cvpn-endpoint-0a06f8f0926c4e211

VPC
vpc-0912600d2a4b66ded

Choose a subnet to associate
subnet-0d9b5f5b35d6b5d02 (private subnet)

Cancel Associate target network

- Eğer çoklu AZ çalışmak istiyorsan birden fazla subnet ekleyebilirsin.

5 Authorization Rule (Yetki Kuralı)

Varsayılan olarak erişim yok.

- Add authorization rule seç.

Client VPN endpoints (1/1) Info

Find client VPN by attribute or tag

Name	Client VPN endpoint ID	State	Client CIDR
client-vpn	cvpn-endpoint-0a06f8f0926c4e211	Pending-associate	172.16.0.0/20

cvpn-endpoint-0a06f8f0926c4e211 / client-vpn

Details Target network associations Security groups **Authorization rules** Route table Connections Tags

Authorization rules Info

Find authorization rule by attribute or tag

Remove authorization rule Add authorization rule

Endpoint ID	State	Description	Group ID	Access all
No authorization rules				
You do not have any authorization rules in this region				

Create authorization rule

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- Target network CIDR: 172.31.0.0/16 (VPC'nin CIDR aralığı)
- Access: "Allow all users" seç.

☰ VPC > Client VPN endpoints > cvpn-endpoint-0a06f8f0926c4e211 > Add authorization rule

Add authorization rule [Info](#)

Add authorization rules to grant clients access to the networks.

Details

Client VPN endpoint ID

cvpn-endpoint-0a06f8f0926c4e211

Destination network to enable access

The IP address, in CIDR notation, of the destination network.

172.31.0.0/16

Grant access to:

☒ Allow access to all users

☐ Allow access to users in a specific access group

Description - optional

A brief description of the authorization rule.

description

Cancel Add authorization rule

6 Route Tanımlama

VPN endpoint üzerinde route ekle:

- Destination: Private subnet CIDR (172.31.0.0/16) (tüm VPC)’ ve “route yerel VPC için otomatik eklenir
- Target: Private subnet

Eğer on-prem ya da başka bir VPC ile Transit Gateway kullanıyorsan, o CIDR’leri de buraya route olarak eklemelisin.

7 Client Config (.ovpn) Dosyası

1. AWS Console → Client VPN Endpoint → Download client configuration.

Successfully initiated addition of rule cvpn-endpoint-0a06f8f0926c4e211.

Client VPN endpoints (1/1) [Info](#)

Find client VPN by attribute or tag

Name	Client VPN endpoint ID	State	Client CIDR
client-vpn	cvpn-endpoint-0a06f8f0926c4e211	Available	172.16.0.0/20

Download client configuration

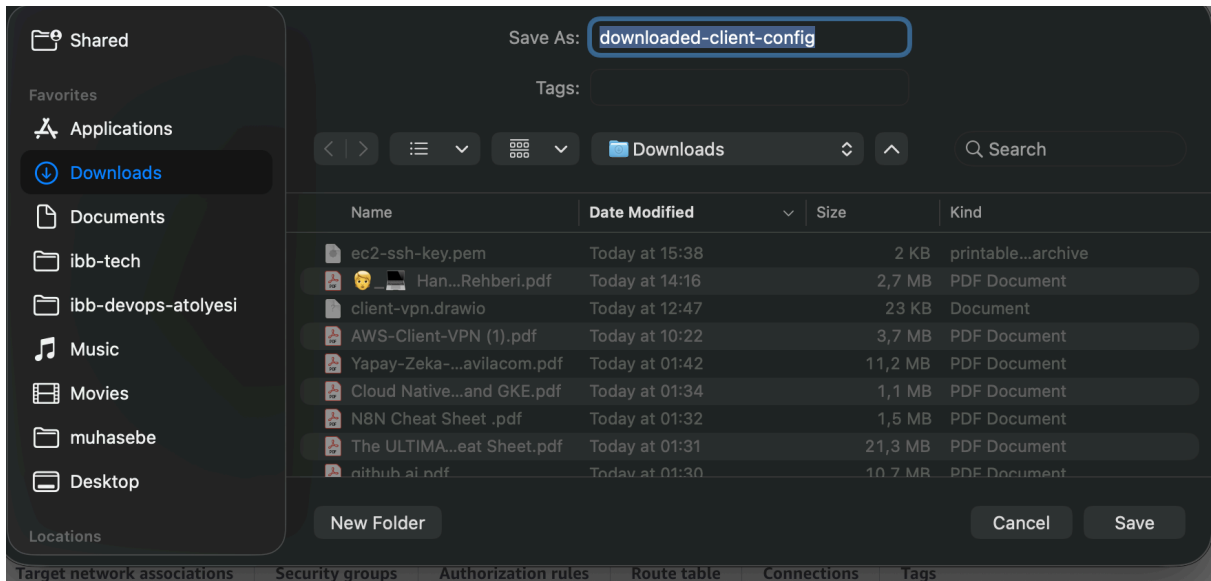
Create client VPN endpoint

Download VPN client configuration file for client VPN endpoint cvpn-endpoint-0a06f8f0926c4e211?

Client VPN endpoint ID

cvpn-endpoint-0a06f8f0926c4e211

Cancel Download client configuration



2. Dosyayı hemen aç ve kontrol et: İçinde aşağıdaki bloklar mutlaka olmalı:

```
<ca>
-----BEGIN CERTIFICATE-----
(... ca.crt içeriği ...)
-----END CERTIFICATE-----
</ca>

<cert>
-----BEGIN CERTIFICATE-----
(... client1.crt içeriği ...)
-----END CERTIFICATE-----
</cert>

<key>
-----BEGIN PRIVATE KEY-----
(... client1.key içeriği ...)
-----END PRIVATE KEY-----
</key>
```

3. Eğer bu bloklar yoksa, indirilen **.ovpn** dosyasının içine kendin eklemen gerekir. Aşağıdaki sertifikaların **.ovpn** dosyasında olması gerekiyor.

pki/ca.crt (Root CA) içeriğini **<ca>** bloğuna koy

pki/issued/client1.devopsatolyesi.com.crt içeriğini **<cert>** bloğuna koy.

pki/private/client1.devopsatolyesi.com.key içeriğini **<key>** bloğuna koy.

Örnek Dosya:

```
client
dev tun
proto udp
```

```
remote
cvpn-endpoint-0a06f8f0926c4e211.prod.clientvpn.us-east-1.amazonaws.com
443
remote-random-hostname
resolv-retry infinite
nobind
remote-cert-tls server
cipher AES-256-GCM
verb 3

<ca>
-----BEGIN CERTIFICATE-----
( buraya pki/ca.crt içeriği gelecek )
-----END CERTIFICATE-----
</ca>

<cert>
-----BEGIN CERTIFICATE-----
( buraya pki/issued/client1.devopsatolyesi.com.crt içeriği gelecek )
-----END CERTIFICATE-----
</cert>

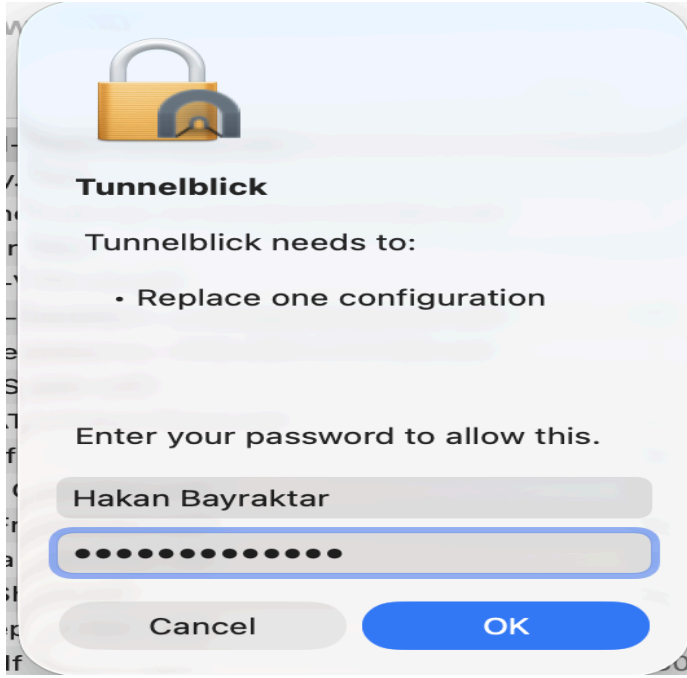
<key>
-----BEGIN PRIVATE KEY-----
( buraya pki/private/client1.devopsatolyesi.com.key içeriği gelecek )
-----END PRIVATE KEY-----
</key>

reneg-sec 0
verify-x509-name devopsatolyesi.com name
```

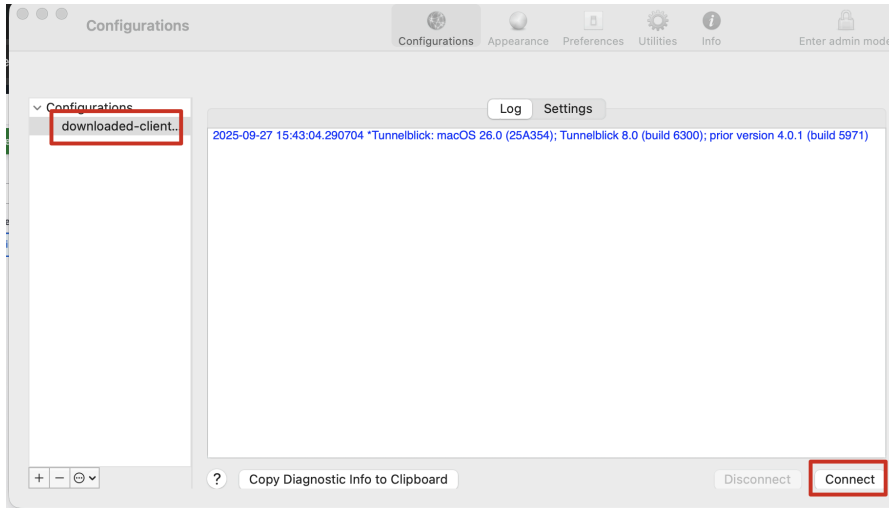
✓ Artık her şey hazır, VPN bağlantısı yapabilirsin.

8) Bağlantı Testi

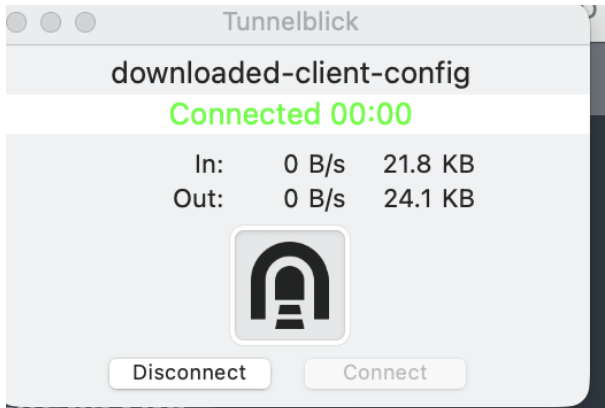
1. **Tunnelblick, Amazon Client VPN for Desktop** veya **OpenVPN Client** indir.
 - **Mac kullanıcıları:** Tunnelblick Download(<https://tunnelblick.net/downloads.html>)
 - **Windows kullanıcıları:** OpenVPN Connect Download(<https://openvpn.net/community-downloads/>)
 - **Mac, Windows, Linux:** Amazon Client VPN for Desktop (<https://aws.amazon.com/vpn/client-vpn-download/>)
2. **.ovpn** dosyasını import et.



3. Connect tıkla → VPN bağlantısı kurulacak.



4. VPN bağlıyken private subnet içindeki EC2'ye SSH ile bağlan:



```
ssh -i mykey.pem ec2-user@172.31.98.215
```

```
^C
(base) aws-client-vpn ➡ ssh -i "ec2-ssh-key.pem" ec2-user@172.31.98.215
The authenticity of host '172.31.98.215 (172.31.98.215)' can't be established.
ED25519 key fingerprint is SHA256:BazeIUTVN1KfoNWHlWLC9o0kIzb6y9f7MYPsVfZwiQ8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.98.215' (ED25519) to the list of known hosts.
```

```
'_#_
~\####_      Amazon Linux 2023
~~~\#####\
~~~~\###|
~~~~\#/      https://aws.amazon.com/linux/amazon-linux-2023
~~~~V~'_->
~~~~
~~~~_. _ .
~~~~/_/_/_/_/_
~~~~/_m/'_/_/_/_/_

[ec2-user@ip-172-31-98-215 ~]$
```

Ec2-test security group için ICMP portlarına izin verelim(ping çekebilmek için)

```
(base) aws-client-vpn ➡ ping 172.31.98.215
PING 172.31.98.215 (172.31.98.215): 56 data bytes
64 bytes from 172.31.98.215: icmp_seq=0 ttl=126 time=136.443 ms
64 bytes from 172.31.98.215: icmp_seq=1 ttl=126 time=136.298 ms
64 bytes from 172.31.98.215: icmp_seq=2 ttl=126 time=136.448 ms
64 bytes from 172.31.98.215: icmp_seq=3 ttl=126 time=136.826 ms
^C
```

- VPN'siz → Private EC2'ye erişim yok.

```
(base) aws-client-vpn ➡ ping 172.31.98.215
PING 172.31.98.215 (172.31.98.215): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
█
```



Lab Sonrası Kaynakları Silme (Clean-up Checklist)

1. **Client bağlantısını kes**
 - Tunnelblick / OpenVPN → Disconnect.
2. **Authorization Rule'ları sil**
 - Client VPN Endpoint → **Authorization** sekmesi → tüm kuralları sil.
3. **Route'ları sil** (eklediysen)
 - Client VPN Endpoint → **Route Table** sekmesi → tüm ek rotaları sil.
4. **Network Association (subnet association) sil**
 - Client VPN Endpoint → **Target Network Associations** → seç → **Disassociate**.
5. **Client VPN Endpoint'i sil**
 - Artık association kalmadığı için "Delete Client VPN Endpoint" başarılı olacak.
6. **EC2'yi terminate et**
7. **ACM sertifikaları** (lab için oluşturduysan) → delete.



Sonuç

Bu yazıda, **AWS Client VPN** servisini kullanarak tamamen AWS yönetiminde, güvenli bir VPN altyapısını nasıl kuracağımızı gördük. Sertifika oluşturma, ACM'e import etme, endpoint tanımlama, subnet association, authorization rule ve route ekleme adımlarını tamamladık. Son olarak client konfigürasyonu ile bağlantı testini yaptık.

Böylece dışarıdan bağlanan kullanıcılar, güvenli şekilde private subnet içindeki kaynaklara erişebilir hale geldi.